Lib-EE

Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [20EE31]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2021

ELECTRICAL CIRCUIT ANALYSIS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Determine the voltages at the nodes in Fig a. Use Cramer's rule. **07**

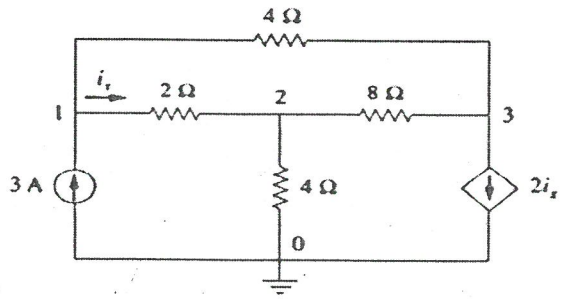

Fig. a.

b. Find the Thevenin equivalent of the circuit in Figure at terminals a-b. **07**

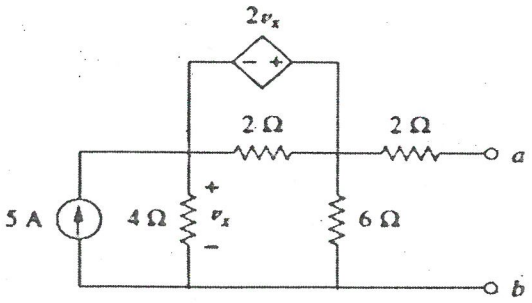

Fig. b.

2. a. Explain the behavior of R, L and C at $t = 0^+$ and at $t = \infty$ with and without initial stored energy. **05**

b. Evaluate the v_c, v_x and i_x for $t > 0$. **05**

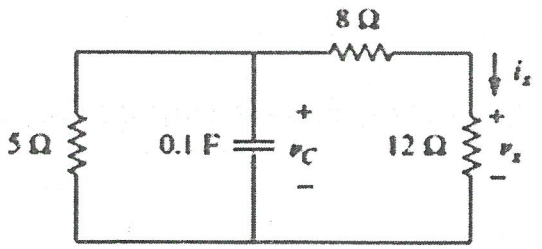

Fig. c.

c. Define the time constants of RL and RC Circuits. What is its significance? **04**

3. **a.** Two identical coils when connected in series across a source, the total inductance is found to be 3.0 H. When one of the coils is reversed the total inductance become 1.4 H. Find the mutual inductance between them and the self-inductance of each coil, Also find the coupling coefficient. **07**

b. An electrical load operates at 240V rms. The load absorbs an average power of 8 kW at a lagging power factor of 0.8. Calculate the impedance and complex power of the load. **07**

OR

4. **a.** A star connected impedance draws a power of 5 kW and a leading current of 100 A from a three phase 415 V, 50 Hz, R,Y,B phase sequence balanced supply. Find the impedance and the circuit elements. Also find the reactive power and power factor. **08**

b. A three-phase unbalanced star connected load is excited by a balanced three phase supply. Derive an expression for the potential difference between the supply neutral and load neutrals. Write expressions for unbalanced phase voltages and also draw the vector diagram. **06**

5. **a.** Find the Laplace transform of the following signals: **08**

i. $f(t) = (2t + 4)u(t)$

ii. $g(t) = (4 + 3e^{-2t})u(t)$

iii. $h(t) = (6\sin(3t) + 8\cos(3t))u(t)$

Determine the inverse Laplace transform of each of the following functions:

iv. $F(s) = \frac{1}{s} + \frac{2}{s+1}$

v. $G(s) = \frac{3s+1}{s+4}$

vi. $H(s) = \frac{12}{(s+2)^2(s+4)}$

b. Calculate the Laplace transform of the periodic function in Fig. d. **06**

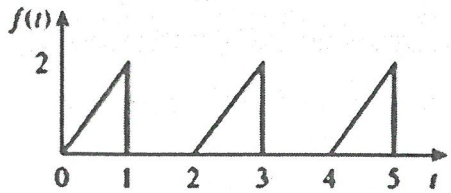

Fig. d.

OR

6. **a.** Determine $y(t) = x(t) * h(t)$ for each paired $x(t)$ and $h(t)$ in Fig. e. to Fig. g. **08**

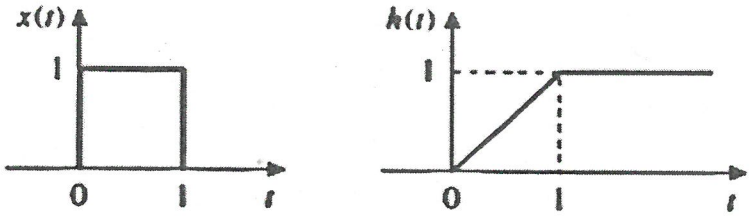

Fig. e.

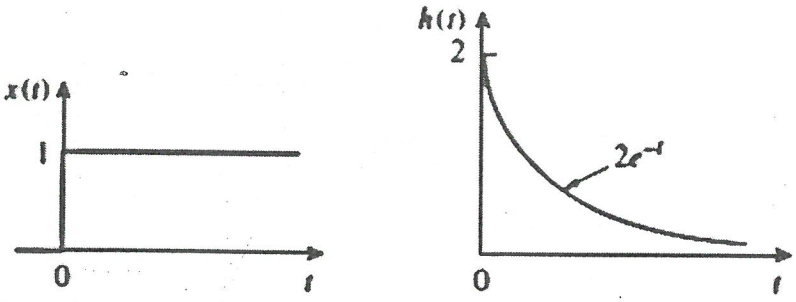

Fig. f.

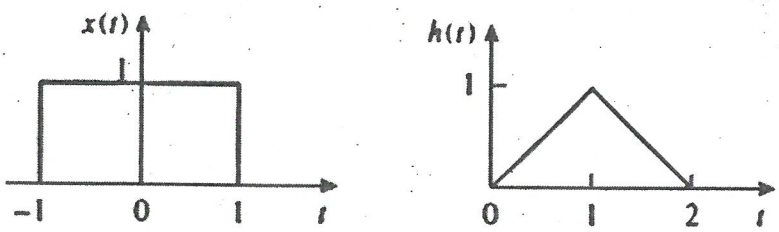

Fig. g.

b. Calculate $v(t)$ given that $V(s) = \frac{10\,s^2+4}{s(s+1)(s+2)^2}$ **06**

7. **a.** Define z and y parameters and obtain the relationship between them. Draw the equivalent circuit of Z – parameters. **06**

b. Determine the *z* parameters for the circuit in Fig. h **04**

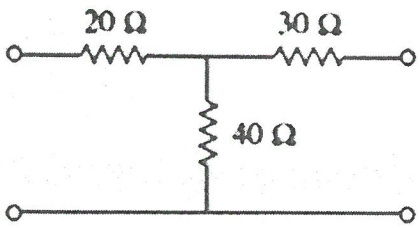

Fig. h.

c. Find $[z]$ and $[g]$ of a two-port network if **04**

$$[T] = \begin{bmatrix} 10 & 1.5\,\Omega \\ 2\,S & 4 \end{bmatrix}$$

OR

8. **a.** Determine the hybrid parameters for the network in Fig. i. **06**

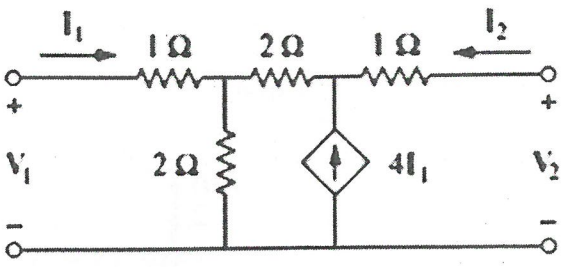

Fig. i.

b. Determine the z and y parameters for the circuit in Fig. j. **04**

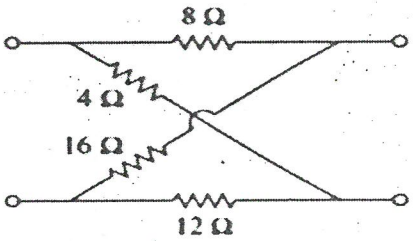

Fig. j.

c. Find the **ABCD** parameters for the circuit in Fig. k. **04**

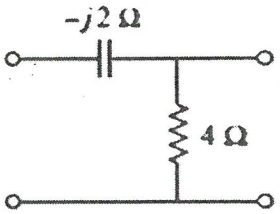

Fig. k.